

(325) Network Design Team – Proposed Scenario 2024

With the recent rise the investments by the US in broadband infrastructure to provide for Internet deployment in all areas of the United States, it has become apparent there is need for more Wireless Internet Service Providers (WISP) to supply the demand of economic growth of internet communication and the advancement of technologies. Increasing the availability of Wireless Internet Service Providers is necessary to meet the requirements of local towns and cities, municipalities, and rural consumers.

As a part of bridging the digital divide and ensuring citizens have access to affordable, reliable, and secure high-speed broadband, it has become apparent that there is a need to build an ISP, primarily a WISP, given the need for flexibility in the network. Approximately 98% of North Dakota can receive 10 Mbps of the Internet through a wired or wireless connection. However, with the demand and need for more bandwidth and availability, the time to enter the market to provide a better solution to the Internet is now.

As a new WISP, it has been decided to provide established services to the rural area of North Dakota just outside of Bismarck. As a startup, it has been agreed to start small and provide a service area of 1,500 square miles surrounding the I-94 corridor between Menoken, ND and Elridge, ND. It is in the works to double or even triple the service area in the future. To niche your company high above the competitors, you will provide high-speed services at and above the 10 Mbps customers can receive currently, at a competitive rate.

The WISP network you design should include high-speed, high-performance, and secure Internet connection to the customers in the area. Your bandwidth should handle Internet streaming, gaming, and the everyday demands consumers need in our ever-changing digital world, as well as the change into providing more value-added service to boost current subscriptions and entice new customers to join.

In addition to constructing the WISP infrastructure, you have built a 2,000-square-foot office building and a 2,000-square-foot metal workshop. Along with developing your ISP wireless network, you must set up a small-office network for your company. Starting with two sales representatives, two marketing representatives, four support agents, two billing/bookkeeping agents, a CEO in the office, and twelve installation team members. Each installation team will be in one truck in pairs of 2. This office and workshop will need a small network with workstations and office equipment to conduct business and support your WISP company customers, and there is a plan to expand the primary office network to remote workers and additional offices as the company grows.

Like any good ISP, you must provide tiered bandwidth services for your customers. The CEO has decided to provide four tiers of service. 1st tier is bandwidth well providing basic internet/email surfing for the residential client, 2nd tier bandwidth provides for Internet streaming for two devices, along with what the 1st tier provides. 3rd tier is for larger families and supports 4K Internet streaming, and includes the tier 1 and 2 features. The 4th tier is the maximum bandwidth your network infrastructure can provide customers without crippling your service provider network while accounting for the customer's needs. In addition to providing internet service, you will supply each customer with an email address and other value-added services as the company progresses.

Needs for your WISP:

- Need to provide a physical solution for a fixed wireless ISP in the planned corridor.
- Provide bandwidth solutions for more than 10 Mbps.
- Need for flexibility allowing expansion to double, even triple the service area in the future.
- Solution for the new office to support the wireless company employees and customers.
- Plan for additional value added services to the network to benefit consumers.
- A solution to support the 4-tier bandwidth delivery packages required by the CEO.

Regional Competition Change

The growth of the company has expanded. Support of the consumers is of the utmost priority. The CEO has decided to take on additional employees in the support department to help with service tickets. However, there is no room in the main office. The CEO has devised a plan to provide remote support agents. Propose a solution to allow support workers to access the company resources and network securely from a remote location.

State Preliminary Competition Change

The CEO has decided to include a small web hosting package allowing 500MBs each to approximately ten recent business clients, in addition to the email service already being provided. Devise a plan to implement a web server to host these small websites within the network.

State Finals Competition Change

Current business subscribers are unhappy with their dated copper PSTN lines supplying service to their landline telephones. In addition, the local telephone company is not repairing broken copper lines forcing their customers to use cell phones. This is not practical for some. As a WISP, a value-added service is to provide VOIP telephone service. VOIP rollout will take some time. Today the CEO wants a planned network segment and be provisioned to account for the additional demand on the network logically. The full rollout of VOIP deployment will come later for the company.

Nationals Preliminary Competition Change

Your company is growing, providing a solution to deploy additional service areas and equipment to another 500 square miles, including other North Dakota cities. As growth continues, the need to formally address a backup/disaster recovery from the unrelenting weather in North Dakota has become apparent to our antennas and towers.

National Finals Competition Change

As a growing WISP company, more value-added services are needed to help boost the subscriptions of the company. In addition to the expanding network of VOIP services recently, you have decided to begin implementing TV streaming. You have agreed to implement 100 of these streaming boxes on a trial basis. Here you will connect to a streaming box attached to a third-party TV provider for local television broadcasting. Devise a plan to implement 100 or more devices within your ISP network to provide TV services to customers.